

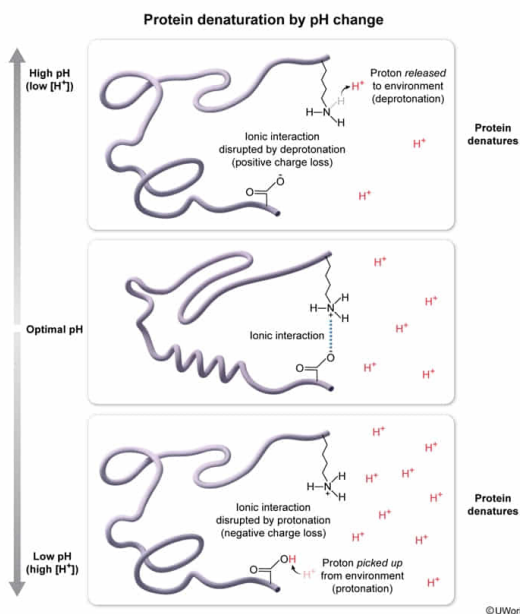
A protein denatures as ionic interactions are disrupted by deprotonation of positively charged side chains. This is most likely the result of:

- ☐ A. a decrease in the ionic strength of the solution. [3%]
- ☐ B. an increase in the ionic strength of the solution. [2%]
- ☐ C. a decrease in the pH of the solution. [16%]
- ✓ ☒ D. an increase in the pH of the solution. [77%]

Correct

76% answered correctly

Explanation:



Proteins are chains of amino acid residues that perform biological functions and, in most cases, must fold into specific three-dimensional structures to perform their functions. Protein folding is driven by the **hydrophobic effect** and **stabilized by interactions between residues** such as **hydrogen bonds** and **ionic bonds**. Ionic interactions take place between positively and negatively charged residues known as **ionizable residues**. **Protein folding is sensitive to the ambient pH** because pH-dependent protonation or deprotonation can *alter* the charges of ionizable amino acids.

By definition, the pH of a solution is equal to $-\log [H^+]$. Thus, at low pH, the concentration of protons (H^+ ions) is high. Accordingly, ionizable amino acids are *more* likely to pick up protons from the environment (ie, become protonated) at low pH. In contrast, at **high pH**, the **proton concentration is low** and ionizable amino acids tend to lose protons to the environment (ie, **become deprotonated**). Protonation can neutralize negative charges, and deprotonation can neutralize positive charges. Either case may **disrupt ionic bonds**.

The question states that ionic bonds in a protein were disrupted by **deprotonation**. Because deprotonation tends to occur in environments with a low concentration of protons (ie, high pH), this result indicates **an increase in the pH of the solution**.

(Choices A and B) Ionic strength is a measure of the concentration of ions in solution. Ions such

as Na^+ or Cl^- can disrupt ionic interactions by **competing** for ionic bonds. Ionic strength does *not* alter the protonation state of amino acid side chains.

(Choice C) A decrease in the pH (ie, *increase* in the ambient proton concentration) of the solution would result in *protonation*, not deprotonation, of ionizable residues.

Educational objective:

The three-dimensional structure of a protein is stabilized by interactions between amino acid residues such as ionic bonds. These ionic bonds can be disrupted by changes in pH, which can alter charges by protonating (decreased pH) or deprotonating (increased pH) ionizable residues.

Time spent: 0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins